Adaptive Retail Advertising System

Additional Interface Control Document + Network Interface Addendum

ICD-5 through ICD-8, plus ICD-NET-1

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Purpose. This document formalizes the four additional interfaces needed to complete the MVP contract surface — Creative Service ↔️ Player, Dashboard Frontend ↔️ Dashboard API, Dashboard API ↔️ PostgreSQL, and Supervisor ↔️ Managed Services — and also formalizes the Orin device's LAN/WAN networking interface. The document is aligned to the locked Golden Build Plan, Roadmap Addendum, Software Selection Directive, and the prior consolidated ICD.

Source basis: Golden Build Plan v1.1; Software Selection Directive; Roadmap Addendum v1; Consolidated ICD v1.0; System Architecture Document.

1. Governing Basis

This addendum inherits the project's locked architectural direction and therefore does not introduce a new system model. It extends the current contract surface so implementation can proceed without schema drift or hidden coupling.

Source Document	Authoritative Guidance Carried Forward
Software Selection Directive	Prefer the best long-term fit, Jetson compatibility, privacy/local-first processing, and clean integration.
Golden Build Plan v1.1	Locks JetPack, Docker, WireGuard, Chromium, DeepStream, FastAPI, PostgreSQL, React, and the practical service layout including creative, dashboard-api, supervisor, and postgres.
Roadmap Addendum v1	Locks adaptive variant rules, dashboard screens, approval controls, and the non-bypassable approval model.
Consolidated ICD v1.0	Provides the base contract style, versioning, and privacy-by-contract approach for metadata-only system interfaces.
System Architecture Document	Requires explicit trust boundaries, interface boundaries, and privacy invariants, including the rule that image data must never be persisted or leave the device.

Normative principles used throughout this document:

- All interface schemas are versioned and backward compatibility rules follow the existing ICD convention.
- All interfaces remain local-first and must operate without WAN connectivity except where remote maintenance or approved update flow explicitly requires it.
- No interface in this document may transmit or persist raw frames, video clips, biometric identifiers, embeddings, or equivalent reversible visual artifacts.

2. Additional Interface Set Overview

Interface	Name	Primary Actors	Primary Transport	Reason it Must Exist
ICD-5	Creative Service	creative, player	HTTPS pull + local	Separates creative

Interface	Name	Primary Actors	Primary Transport	Reason it Must Exist
	↔️ Player		file cache	assembly from rendering and prevents frontend/backend coupling.
ICD-6	Dashboard Frontend ↔️ Dashboard API	dashboard UI, dashboard-api	HTTPS / REST	Formalizes the business-facing control plane and approval workflow.
ICD-7	Dashboard API ↔️ PostgreSQL	dashboard-api, postgres	SQLAlchemy / PostgreSQL	Locks canonical persistence, migrations, and authoritative schemas.
ICD-8	Supervisor ↔️ Managed Services	supervisor, service containers	HTTP health + local control actions	Formalizes self-healing, maintenance windows, and restart semantics.
ICD-NET-1	Orin Device ↔️ LAN/WAN Network	Orin device, LAN, VPN admin node	Ethernet/Wi-Fi, WireGuard	Locks exposure model, VPN-only remote access, and WAN-independent runtime operation.

3. Cross-Cutting Rules

Rule	Requirement
Schema versioning	Every payload and entity contract SHALL carry schema_version. v1.0 is the baseline for this addendum.
Time format	All timestamps SHALL be UTC RFC3339 strings.

Rule	Requirement
Correlation	Every externally observable event SHALL carry a correlation_id or equivalent request_id/command_id.
Privacy flags	All relevant payloads SHALL explicitly assert that no images, frame URLs, or face embeddings are present where applicable.
Auth posture	No interface in this document SHALL rely on public unauthenticated WAN exposure.
Failure posture	If a dependent interface fails, the system SHALL degrade to a safe local fallback and the screen SHALL never go blank.

4. ICD-5 — Creative Service ↔ Player

Primary purpose: Deliver deterministic, approved render instructions from the creative service to the player so the player can activate a local creative without embedding business logic or freeform generation logic in the frontend.

Field	Value
Producer	creative service
Consumer	player
Transport	HTTPS on LAN/VPN, plus local asset cache access on the device
Primary operation	Player retrieves a CreativeRenderManifest for a referenced creative_id/assembly_id
Privacy posture	No audience identity data; no camera data; no generated content outside approved policy space

4.1 Contracted Objects

The contract is split into a lightweight command reference and a deterministic render manifest. ICD-4 remains the authoritative decision-to-player command interface. ICD-5 defines the manifest the player renders after a valid activate_creative command.

4.2 CreativeRenderManifest Schema


```
{
  "schema_version": "1.0",
  "message_type": "creative.render_manifest",
  "manifest_id": "uuid",
  "produced_at": "2026-03-17T12:00:00Z",
  "tenant_id": "tenant-acme",
  "site_id": "site-001",
  "creative_id": "creative-spring-sale-15s",
  "assembly_id": "assembly-00017",
  "template": {
    "template_id": "promo_split_v1",
    "template_version": "3"
  },
  "slots": {
    "headline": {
      "kind": "text",
      "component_id": "comp-headline-01",
      "value": "Fresh Coffee Daily"
    },
    "body": {
      "kind": "text",
      "component_id": "comp-body-09",
      "value": "Order at the counter and save this week."
    },
    "hero": {
      "kind": "asset",
      "asset_id": "asset-img-102",
      "checksum": "sha256:..."
    },
    "cta": {
      "kind": "text",
      "component_id": "comp-cta-02",
      "value": "Order Now"
    }
  },
  "style": {
    "palette_id": "brand-warm-1",
    "background": "asset-bg-011",
    "font_pack": "font-pack-default",
    "accent_hex": "#FF9900"
  },
  "policy": {
    "approved": true,
    "approval_mode": "manual",
    "contains_generated_text": true,
    "contains_generated_image": false,
    "audio_allowed": false
```

```
    },
    "render_constraints": {
      "duration_ms": 15000,
      "transition": "cut",
      "mute": true,
      "offline_cache_required": true
    }
  }
```

4.3 Normative Rules

- The player SHALL render only manifests where policy.approved = true.
- The player SHALL reject manifests that reference uncached required assets when offline_cache_required = true and SHALL send a rejection acknowledgement with reason = CACHE_MISS.
- The creative service SHALL only emit text or style variants that conform to the approved modifier policy; it SHALL NOT emit prompt-generated images or unconstrained layouts.
- The player SHALL treat the manifest as deterministic render data, not as HTML content to be executed from an untrusted source.
- The player SHALL support a local fallback creative bundle if the creative service is unavailable.

4.4 Failure and Recovery Semantics

Condition	Required Behavior
Manifest missing	Player continues current ad until timeout threshold, then enters approved fallback playlist.
Unapproved manifest	Reject and log policy violation; never render.
Asset checksum mismatch	Reject manifest and request safe fallback.
Creative service unavailable	Use last valid manifest if still within policy; otherwise use fallback bundle.

4.5 Observability

- Emit CreativeFetchEvent, CreativeRenderEvent, CreativeRejectEvent, and AssetCacheMissEvent.
- Track manifest fetch latency, render success rate, asset cache hit rate, and creative rejection reasons.

5. ICD-6 — Dashboard Frontend ↔ Dashboard API

Primary purpose: Define the business-facing API contract used by the local dashboard for campaigns, creative approvals, analytics, system status, and settings. This interface is the owner/admin control plane and SHALL enforce the locked approval model and manual override rules.

Field	Value
Producer/Caller	dashboard frontend (React)

Field	Value
Consumer/Server	dashboard-api (FastAPI)
Transport	HTTPS on LAN/VPN only
Auth	Session or bearer-token auth on LAN/VPN; no public WAN access
Authority	dashboard-api remains canonical data model authority

5.1 Endpoint Families

Endpoint Family	Purpose	Minimum Methods
/v1/campaigns	Create, edit, enable, disable campaigns	GET, POST, PATCH
/v1/assets	Upload and inspect approved assets	GET, POST
/v1/approvals	Approve or reject pending assets/variants	GET, POST
/v1/analytics	Read local analytics and experiment results	GET
/v1/system	Read health, service state, WireGuard status, maintenance window	GET, PATCH
/v1/settings	Update device and operational settings	GET, PATCH

5.2 Representative Request/Response Objects

```
{
  "schema_version": "1.0",
  "message_type": "approval.action.request",
  "request_id": "uuid",
  "actor_id": "user-001",
  "actor_role": "owner",
  "approval_target_type": "variant",
```

```
  "approval_target_id": "variant-145",
  "action": "approve",
  "comment": "Approved for trial in campaign spring-promo",
  "effective_at": "2026-03-17T20:00:00Z"
}
{
  "schema_version": "1.0",
  "message_type": "campaign.upsert.request",
  "campaign_id": "campaign-spring-2026",
  "name": "Spring Counter Promo",
  "status": "enabled",
  "template_ids": [
    "promo_split_v1"
  ],
  "variant_group_ids": [
    "vg-001",
    "vg-003"
  ],
  "start_at": "2026-03-18T07:00:00Z",
  "end_at": "2026-03-31T23:00:00Z"
}
```


5.3 Normative Rules

- The API SHALL enforce the non-bypassable approval system: no unapproved asset, text variant, image variant, or modifier output may be marked runtime-eligible via a dashboard call.
- Manual override controls SHALL always exist and SHALL permit safe mode / fallback playlist activation without requiring WAN connectivity.
- Dashboard analytics responses SHALL expose aggregate and explainable metrics only; they SHALL NOT surface raw CV imagery or person-level identifiers.
- The API SHALL treat WireGuard status, maintenance window, and safe mode playlist as managed settings that can be read in Settings/System Status views.

5.4 Errors and Idempotency

Status	Use
400	Invalid payload or schema violation
401/403	Missing or insufficient authentication/authorization
404	Referenced campaign/asset/variant not found
409	State conflict, such as approving an already rejected object or enabling a campaign with no approved assets
422	Business rule violation within otherwise valid request

Mutating requests SHOULD use request_id for idempotent retry handling.

5.5 Observability

- Emit ApprovalActionEvent, CampaignChangeEvent, SettingsChangeEvent, and DashboardSessionEvent.
- Track API p50/p95 latency, approval queue depth, failed mutations, and safe-mode activations initiated through the dashboard.

6. ICD-7 — Dashboard API ↔ PostgreSQL (Canonical Persistence Interface)

Primary purpose: Formalize canonical persistence so service logic, analytics, and approvals do not drift into ad hoc storage patterns. This interface is internal to the device but is treated as a versioned contract because persistence changes create the highest hidden coupling risk.

Field	Value
Producer/Caller	dashboard-api (and only dashboard-api for canonical writes in MVP)
Consumer	postgres
Transport	Local PostgreSQL connection via SQLAlchemy
Auth	Local DB credentials mounted at runtime as secrets
Write authority	dashboard-api is the sole canonical schema authority for MVP writes

6.1 Canonical Entity Set

Entity	Purpose	Write Rule
Campaign	Runtime business grouping of approved creatives	Writable only via dashboard-api
Template / SlotDefinition	Creative layout definitions	Writable only via dashboard-api
Asset / CreativeComponent	Approved content primitives	Writable only via dashboard-api
VariantGroup / ModifierPolicy	Allowed adaptive creative scope	Writable only via dashboard-api
CreativeAssembly / AdVariant	Materialized render-ready creative definitions	Writable only via dashboard-api or creative service through dashboard-api-owned workflow

Entity	Purpose	Write Rule
AudienceState / DecisionEvent / AdPlaybackEvent	Operational and analytic records	Append-only runtime events
SystemFaultEvent / MaintenanceEvent	Reliability and audit records	Append-only runtime events

6.2 Persistence Rules

- SQLite SHALL NOT be used as the canonical production store.
- Schema changes SHALL be applied only through versioned migrations.
- Runtime event tables SHALL prefer append-only semantics; mutable business entities SHALL maintain updated_at and version fields.
- Audience and analytics persistence SHALL remain aggregate and metadata-only; no raw frames, embeddings, or person-level identifiers may be stored.
- Database unavailability SHALL NOT blank the screen; playback continues and minimal logs may be buffered until persistence returns.

6.3 Representative Canonical Table Shape

```
{
  "table": "ad_playback_event",
  "primary_key": "playback_event_id",
  "columns": {
    "schema_version": "text not null",
    "tenant_id": "text not null",
    "site_id": "text not null",
    "device_id": "text not null",
    "creative_id": "text not null",
    "assembly_id": "text",
    "campaign_id": "text",
    "started_at": "timestamptz not null",
    "ended_at": "timestamptz",
    "result": "text not null",
    "decision_event_id": "text",
    "correlation_id": "text not null"
  }
}
```

6.4 Transaction and Migration Semantics

- Business object mutations SHALL occur within explicit transactions and SHALL fail atomically.
- Migrations SHALL be forward-only in normal operation, with documented rollback plans for release failures.
- Application code SHALL not assume it can auto-create production schema on first boot; schema bootstrap is part of the controlled provisioning path.

6.5 Retention and Audit

Data Class	Minimum Rule
Business entities	Retain until deleted/archived by policy or operator action
Decision and playback events	Retain long enough to compare static versus adaptive performance during pilot; exact retention window is deployment-configurable
Fault and maintenance events	Retain for operational audit and troubleshooting
Any visual data	Retention prohibited

7. ICD-8 — Supervisor ↔ Managed Services

Primary purpose: Define how the supervisor monitors, classifies, and acts on service health so the runtime self-heals without violating the conditional maintenance policy or the rule that the screen must never go blank.

Field	Value
Producer/Controller	supervisor service
Consumers	input-cv, audience-state, decision-optimizer, creative, player, dashboard-api, postgres
Transport	HTTP health probes, container runtime inspection, local control actions
Primary actions	observe, classify, restart, restart subsystem, request safe mode, schedule conditional reboot
Policy source	Golden Build Plan recovery ladder and maintenance-window rules

7.1 Required Managed Service Endpoints

Service	Minimum Endpoint / Signal	Required Output
player	/healthz and heartbeat event	ready state, current creative, last ack age

Service	Minimum Endpoint / Signal	Required Output
input-cv	/healthz and camera status	camera online/offline, frame age, decoder status
audience-state	/healthz	last input age, last state_seq, stable/unstable
decision-optimizer	/healthz	last decision age, queue depth
creative	/healthz	cache state, manifest generation status
dashboard-api	/healthz	API ready, DB reachable
postgres	connection probe	read/write availability

7.2 SupervisorControlEvent Schema

```
{
  "schema_version": "1.0",
  "message_type": "supervisor.control_event",
  "event_id": "uuid",
  "produced_at": "2026-03-17T12:00:00Z",
  "target_service": "player",
  "health_state": "degraded",
  "action": "restart_service",
  "reason": "heartbeat_timeout",
  "maintenance_window": "02:00-04:00",
  "requires_operator_notice": false
}
```

7.3 Normative Rules

- The supervisor SHALL attempt restart failing container/service first, then subsystem restart, then conditional off-hours reboot only when thresholds justify it.
- The supervisor SHALL NOT perform blind nightly or mandatory daily reboots.
- If the player becomes unhealthy, the supervisor SHALL prefer restoring the player process or forcing fallback playlist mode before any device reboot path.
- If dashboard-api is unavailable, playback SHALL remain unaffected.
- If postgres is unavailable, the supervisor SHALL preserve visible runtime function and classify persistence degradation separately from playback failure.

7.4 Failure Classification Matrix

Class	Examples	Required System Behavior
Playback critical	player crash, repeated command ack timeout	Recover player first; maintain visible fallback; never blank

Class	Examples	Required System Behavior
		screen
CV degraded	camera offline, decoder stall	Fallback rotation; mark audience-dependent decisions degraded
Persistence degraded	DB unavailable	Continue playback; buffer minimal logs where feasible
Admin degraded	dashboard unavailable, VPN unavailable	Local playback continues; remote support impaired only

8. ICD-NET-1 — Orin Device ↔ LAN/WAN Network

Primary purpose: Lock the device's exposure model, network trust boundaries, allowed traffic classes, and remote-operations path so deployment does not drift into insecure or cloud-dependent patterns.

Field	Value
Device role	Local-first edge appliance attached to customer LAN
LAN requirement	Required for dashboard access and optional NTP/DNS reachability
WAN requirement	Not required for runtime operation; only required for approved maintenance/update paths
Inbound WAN	Prohibited
Remote admin path	WireGuard only

8.1 Trust Boundaries

- Trusted device boundary: the Orin host OS, Docker networks, and local storage.
- Local network boundary: customer LAN; dashboard may be exposed here under local auth policy.
- Remote management boundary: WireGuard VPN overlay only.
- Public WAN boundary: untrusted; no public service exposure is allowed.

8.2 LAN Exposure Contract

Service	Host Exposure	Access Scope	Notes
dashboard-ui	yes	LAN + VPN	Business-facing local interface
dashboard-api	yes	LAN + VPN	Backend for dashboard only
metrics endpoint	optional	VPN or restricted LAN	Operational use only
postgres	no	internal only	Never directly exposed
input-cv / audience-state / decision / creative / player control	no	internal only	Bound to internal Docker networking

8.3 WAN/Egress Contract

Traffic Class	Allowed	Reason
WireGuard tunnel establishment	Yes	Remote administration and controlled support
NTP / DNS	Yes	Time and name resolution where available
Manual update retrieval	Yes	Controlled off-hours updates during pilot
Cloud runtime dependency	No	Core system must remain WAN-independent
Camera/image data egress	No	Never leaves device

8.4 Addressing and Bootstrap

- Default network mode SHALL be DHCP with static lease recommended at the customer router.
- Hostname SHOULD follow signage-{device_id}.
- VPN bootstrap SHALL provision per-device keys; SSH, if enabled, SHALL be reachable only over VPN.
- No exposed SSH without VPN is permitted.

8.5 Firewall and Exposure Rules

- Default policy: deny inbound WAN; allow outbound WAN for approved classes only.
- Default policy: expose only dashboard-ui and dashboard-api to LAN; all runtime services remain internal.

- Reverse proxies and logs SHALL redact query strings if short-lived access tokens are ever used in WebSocket URLs or similar local interfaces.

8.6 Failure and Degraded Operation

Condition	Required Behavior
LAN unavailable	Player continues current/fallback operation locally; dashboard unreachable
WAN unavailable	Normal runtime continues; remote support and manual update retrieval unavailable
WireGuard down	Local owner functions remain; remote admin unavailable
DNS/NTP unavailable	System uses cached config/time strategy where feasible; no effect on visible playback path

8.7 Network Observability

- Track VPN state, device IP, dashboard reachability, DNS/NTP reachability, and failed remote admin attempts.
- Do not collect or export traffic captures containing visual data; network telemetry remains operational metadata only.

9. Versioning, Compatibility, and Open Items

Topic	Rule
Schema compatibility	Patch releases may clarify behavior only; minor releases may add optional fields; major releases may break shape or semantics.
Transport changes	Changing transport without changing payload semantics SHALL still require an ICD revision and conformance testing.
Current TBDs	Identity provider details, exact dashboard auth mechanism, exact LAN port assignments, and exact event retention windows remain deployment-defined.

10. Implementation Checklist

- Implement Pydantic/OpenAPI schemas for ICD-5 and ICD-6 request/response objects.
- Generate SQLAlchemy models and migrations consistent with ICD-7 entity rules.
- Implement supervisor health endpoints and restart event schema from ICD-8.
- Bind only dashboard-ui and dashboard-api to the host; keep remaining services on internal Docker networks.
- Provision WireGuard before enabling remote administration.
- Add conformance tests for unapproved creative rejection, approval workflow invariants, DB migration integrity, service-restart logic, and WAN-independent playback operation.

11. Changelog

v1.0 - 2026-03-17 - Initial formalization of five additional interfaces: Creative ↔️ Player, Dashboard Frontend ↔️ Dashboard API, Dashboard API ↔️ PostgreSQL, Supervisor ↔️ Managed Services, and Orin ↔️ LAN/WAN Network.